PROPOSAL KERJA PRAKTIK

PT. JATIM AUTOCOMP INDONESIA



Oleh:

Muhammad Zaki Zain F.	3323600006
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Daniar Priambodo	3323600009
Zulfahmi Nafiis Suffi	3323600019

PROGRAM STUDI D4 SAINS DATA TERAPAN
JURUSAN TEKNIK INFORMATIKA DAN KOMPUTER
POLITEKNIK ELEKTRONIKA NEGERI SURABAYA
TAHUN AJARAN 2025-2026

LEMBAR PENGESAHAN
PROPOSAL KERJA PRAKTIK
PT JATIM AUTOCOMP INDONESIA

Pemohon 1	:	Muhammad Zaki Zain Fanuruddin Putra	NRP. 3323600006	.....
Pemohon 2	:	Zacky Ainul Mustofa	NRP. 3323600007	.....
Pemohon 3	:	Daniar Priambodo	NRP. 3323600009	.....
Pemohon 4	:	Zulfahmi Nafiis Suffi	NRP. 3323600019	.....

Mengetahui,

Koordinator Program Studi
D4 Sains Data Terapan PENS

Isbat Uzzin Nadhori S.Kom., M.T
NIP. 197405052003121002

Koordinator Kerja Praktik
D4 Sains Data Terapan PENS

Alfi Fadliana S.Si., M.Stat NIP. 199211252022032018

BAB 1
PENDAHULUAN

1.1 LATAR BELAKANG

Di era Revolusi Industri 4.0, industri manufaktur mengalami transformasi besar-besaran menuju smart manufacturing yang mengintegrasikan teknologi digital, Internet of Things (IoT), dan analisis data untuk meningkatkan efisiensi produksi. Data telah menjadi aset strategis yang dapat dimanfaatkan untuk optimalisasi proses, peningkatan kualitas produk, prediksi kebutuhan maintenance, dan pengambilan keputusan yang lebih akurat dan cepat.

PT Jatim Autocomp Indonesia (JAI) merupakan perusahaan manufaktur komponen otomotif yang berlokasi di Gempol, Pasuruan, yang berperan penting dalam rantai pasok industri otomotif nasional. Sebagai perusahaan manufaktur modern, PT JAI menghadapi berbagai tantangan operasional seperti optimalisasi proses produksi, pengendalian kualitas produk, efisiensi penggunaan mesin dan material, serta prediksi permintaan pasar. Tantangan-tantangan ini memerlukan pendekatan berbasis data yang sistematis dan terukur. Mahasiswa Program Studi Sains Data Terapan dididik untuk memiliki kompetensi teknis dalam mengolah data, mulai dari pengumpulan data, pembersihan data, eksplorasi data, pembuatan model statistik maupun machine learning, hingga pembuatan visualisasi data interaktif.

Penerapan ilmu Sains Data di lingkungan manufaktur dapat memberikan nilai tambah signifikan, seperti predictive maintenance untuk mengurangi downtime mesin, quality control berbasis machine learning untuk mendeteksi defect produk secara dini, analisis efisiensi produksi untuk mengidentifikasi bottleneck proses, serta demand forecasting untuk optimalisasi inventory. Dengan memanfaatkan data historis produksi, data sensor mesin, dan data quality control, berbagai permasalahan operasional dapat dianalisis dan diselesaikan dengan pendekatan yang lebih objektif dan terukur.

Melalui kerja praktek ini, kami bermaksud mengimplementasikan teknik-teknik sains data terapan untuk membantu PT JAI dalam mengelola dan menganalisis data operasional dan produksi. Diharapkan, sinergi ini dapat menghasilkan solusi berbasis data berupa dashboard monitoring produksi, model prediktif untuk quality control atau maintenance, atau analisis optimalisasi proses yang dapat mendukung peningkatan produktivitas dan daya saing perusahaan di industri komponen otomotif.

1.2 TUJUAN

Berdasarkan latar belakang tersebut, tujuan pelaksanaan Kerja Praktek ini adalah:

1. Memenuhi persyaratan kurikulum Program Studi Sains Data Terapan di Politeknik Elektronika Negeri Surabaya
2. Merasakan pengalaman nyata bekerja sebagai Data Analyst/Data Scientist di lingkungan industri manufaktur komponen otomotif
3. Mengimplementasikan ilmu yang telah didapat selama perkuliahan baik teori maupun praktikum dalam konteks industri manufaktur yang sesungguhnya
4. Menekuni serta memperkuat keterampilan yang dimiliki pada bidang sains data terapan, khususnya dalam analisis data produksi, predictive modeling, dan data visualization
5. Memberikan kontribusi nyata kepada PT JAI melalui solusi berbasis data yang dapat meningkatkan efisiensi operasional, kualitas produk, atau produktivitas perusahaan

1.3 MANFAAT

a. Bagi Mahasiswa

1. Dapat memenuhi tugas Kerja Praktek sebagai syarat wajib kelulusan mahasiswa Diploma IV Politeknik Elektronika Negeri Surabaya
2. Memberikan nilai tambah berupa wawasan, pengetahuan, dan pengalaman sebagai generasi muda yang nantinya akan berkontribusi dan terjun langsung pada dunia kerja, khususnya di sektor manufaktur
3. Mengetahui secara langsung kendala dan permasalahan yang terjadi di industri manufaktur serta bagaimana solusi penyelesaiannya menggunakan pendekatan berbasis data
4. Memperoleh pengalaman kerja dan mengetahui bagaimana penerapan sains data di industri manufaktur komponen otomotif
5. Membangun portfolio project berbasis industri yang dapat memperkuat daya saing di dunia kerja

b. Bagi Program Studi Sains Data Terapan Politeknik Elektronika Negeri Surabaya

1. Menjadi tolak ukur ilmu sains data terapan secara pembelajaran dalam peningkatan kualitas pengajaran di masa yang akan datang
2. Sebagai sarana pengenalan dan pembelajaran yang efektif bagi mahasiswa sebelum lulus dan terjun pada dunia kerja nyata, khususnya di sektor industri manufaktur
3. Sarana mengenal kualitas mahasiswa Politeknik Elektronika Negeri Surabaya pada perusahaan manufaktur komponen otomotif

4. Memperkuat kerja sama dengan dunia industri manufaktur dalam pengembangan kurikulum yang relevan dengan kebutuhan industri

c. Bagi PT Jatim Autocomp Indonesia (JAI)

1. Membangun pengetahuan pada para mahasiswa untuk mengetahui proses kerja dan operasional di industri manufaktur komponen otomotif
2. Memberikan peran dan kontribusi positif kepada masyarakat dan mahasiswa khususnya dalam membangun dan meningkatkan kualitas sumber daya manusia Indonesia sebagai generasi muda penerus bangsa
3. Sebagai sarana media untuk meningkatkan kerjasama antara PT Jatim Autocomp Indonesia dengan Politeknik Elektronika Negeri Surabaya
4. Mendapatkan perspektif fresh dan solusi inovatif berbasis data science yang dapat mendukung optimalisasi proses bisnis dan operasional perusahaan
5. Memperoleh dukungan dalam proyek-proyek analisis data yang dapat meningkatkan efisiensi produksi, kualitas produk, atau produktivitas

BAB 2
PELAKSANAAN

2.1 BENTUK KEGIATAN

Selama kerja praktek di PT Jatim Autocomp Indonesia (JAI), mahasiswa akan mengikuti serangkaian kegiatan yang mengintegrasikan teori dan praktik langsung yang mencakup berbagai aktivitas yang disesuaikan dengan bidang keilmuan yang ditekuni, yaitu Sains Data Terapan. Fokus utama kegiatan diarahkan pada pengamatan, analisis, serta penerapan teknologi dan metode pengolahan data yang mendukung proses operasional dan produksi di lingkungan manufaktur komponen otomotif.

Kegiatan meliputi orientasi dan pemahaman proses produksi serta sistem yang berjalan di PT JAI, identifikasi kebutuhan analisis data pada area produksi, quality control, maintenance, atau supply chain, pengumpulan dan pembersihan data dari berbagai sumber seperti sistem ERP, sensor mesin, atau database produksi, exploratory data analysis (EDA) untuk memahami pola dan karakteristik data operasional, pengembangan model analisis menggunakan teknik statistik dan machine learning untuk predictive maintenance, quality control, atau optimalisasi proses, pembuatan visualisasi data dan dashboard interaktif untuk monitoring real-time, serta dokumentasi dan penyusunan laporan akhir.

Dengan demikian, kegiatan ini diharapkan dapat memberikan kontribusi nyata terhadap upaya peningkatan efisiensi operasional berbasis sains data di PT JAI, sekaligus memperkaya pemahaman mahasiswa terhadap penerapan ilmu data dalam konteks industri manufaktur komponen otomotif.

2.2 WAKTU PELAKSANAAN KEGIATAN

Kegiatan kerja praktik akan dilaksanakan selama 6 (enam) bulan, dimulai pada tanggal 5 Januari 2026 dan berakhir pada tanggal 29 Juni 2026. Periode pelaksanaan ini mengacu pada Peraturan Akademik Program Studi Sarjana Terapan Sains Data Terapan Politeknik Elektronika Negeri Surabaya Tahun Akademik 2025/2026 dan disesuaikan dengan kalender akademik yang berlaku.

2.3 LOKASI KEGIATAN

Kegiatan kerja praktik akan dilaksanakan di PT Jatim Autocomp Indonesia (JAI), yang beralamat di Jl. Raya Wonoayu No. 26, Gempol, Pasuruan, Jawa Timur.

2.1 RENCANA KEGIATAN

Adapun rencana kegiatan selama kerja praktek adalah sebagai berikut:

No	Kegiatan	Bulan																							
		Januari				Februari				Maret				April				Mei				Juni			
		1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4
1	Orientasi dan Pemahaman Sistem Produksi Wiring Harness																								
2	Pengumpulan Data Produksi, Defect, dan Quality Control																								
3	Data Cleaning dan Preprocessing																								
4	Analisis Defect Pattern dan Trend Produksi																								
5	Pengembangan Model Predictive Quality Control																								
6	Pembuatan Dashboard Monitoring Line Produksi																								
7	Testing dan Validasi Dashboard																								
8	Penyusunan Laporan Kerja Praktik																								

2.2 LINGKUP MATERI

Beberapa ilmu yang telah diperoleh serta mendukung pelaksanaan kegiatan Kerja Praktik ini adalah:

- Statistika Dasar
- Pemodelan Statistik Terapan
- Analisa Statistika Terapan
- Analitika Data Terapan
- Pemrograman 1, 2, dan 3
- Basis Data
- Data Warehouse
- Pemrosesan Data
- Manajemen Data
- Eksplorasi dan Visualisasi Data
- Teknik Presentasi Data
- Matematika 1, 2, dan 3
- Text Mining
- Data Mining
- Aljabar Linier
- Logika dan Algoritma
- Kecerdasan Buatan
- Mesin Pembelajaran
- Neuro Computing
- Machine Learning Ops
- Teknologi Web Service
- Infrastruktur dan Manajemen Big Data
- Teknologi dan Tool Big Data
- Ekonometrika Terapan
- Technopreneur Sains Data
- Bahasa Inggris Teknik
- Bahasa Indones

2.3 SASARAN KEGIATAN

Sasaran dari kegiatan Kerja Praktik ini adalah:

1. Peserta Kerja Praktik / Mahasiswa
2. PT Jatim Autocomp Indonesia (JAI)
3. Pembimbing Kerja Praktik (Perusahaan/Kampus)

2.4 PESERTA

Peserta yang mengikuti kerja praktik ini merupakan mahasiswa Program Studi Sarjana Terapan Sains Data Terapan Politeknik Elektronika Negeri Surabaya, dengan identitas sebagai berikut:

NAMA	NRP
Muhammad Zaki Zain Fanuruddin Putra	3323600006
Zacky Ainul Mustofa	3323600007
Daniar Priambodo	3323600009
Zulfahmi Nafiis Suffi	3323600019

BAB 3
PENUTUP

Demikian proposal permohonan Kerja Praktik ini kami susun sebagai kerangka acuan dan gambaran singkat mengenai kegiatan yang akan dilaksanakan di PT Jatim Autocomp Indonesia (JAI).

Kami menyadari bahwa proposal ini masih bersifat rencana yang fleksibel. Oleh karena itu, kami sangat terbuka terhadap segala bentuk penyesuaian, bimbingan, serta arahan dari pihak PT JAI terkait topik, teknis pelaksanaan, maupun jadwal kegiatan agar selaras dengan kebutuhan dan prioritas perusahaan.

Besar harapan kami untuk dapat diberikan kesempatan bergabung dan berkontribusi tenaga serta pikiran, khususnya dalam penerapan ilmu Sains Data Terapan untuk mendukung optimalisasi proses produksi dan operasional di PT JAI. Kami siap mematuhi segala peraturan dan tata tertib yang berlaku, serta menjaga nama baik almamater dan perusahaan selama pelaksanaan kegiatan ini. Atas perhatian, kesempatan, dan kerja sama Bapak/Ibu, kami ucapkan terima kasih

LAMPIRAN PEMOHON 1
CURRICULUM VITAE



Daniar Priambodo

Address : Semeru Street, Lirboyo, Mojoroto, Kediri City, East Java
Phone : +6288217832447
Email : daniarpriambodo11@gmail.com
Linkedin : https://www.linkedin.com/in/daniar-priambodo-2a9337363/

PROFIL

I'm an Applied Data Science undergraduate who enjoys turning data into clear, meaningful insights. I've worked on several analytics and dashboard projects using Python and modern visualization tools like Tableau, Grafana, and Power BI. I adapt quickly, collaborate well in teams, and am currently seeking an internship where I can grow while contributing to impactful data projects.

EDUCATION

Electronic Engineering Polytechnic Institute of Surabaya **Aug 2023 - Present**
Applied Data Science, GPA 3,51/4.00

Project Experience

SiDesa - Analysis of Disparities and Typologies of Village Development in East Java
Data Science / Data Analyst **Aug 2025 – Des 2025**

- Built an interactive spatial dashboard to visualize regional disparities.
- Applied ML clustering (K-Means, Hierarchical) to classify village development typologies.
- Processed and integrated socioeconomic data with GIS shapefiles for geospatial analysis.

Langit.in – Climate Forecasting System for East Java
Data Science / Data Analyst **Feb 2025 – Jun 2025**

- Developed climate forecasting models using time-series analysis and machine learning techniques.
- Processed and cleaned multi-year climate datasets to generate accurate forecasting inputs.
- Evaluated model performance using RMSE, MAE, and cross-validation to ensure prediction accuracy.

ERP Supply Chain Data Mart – Production Module Integration
Data Engineer **May 2025 – Jun 2025**

- Developed and managed ETL pipelines using Pentaho Data Integration (PDI) to integrate production module data into the Supply Chain Data Mart.
- Performed data extraction from PostgreSQL and MySQL ERP sources, followed by transformation and loading into centralized analytical tables.
- Designed optimized fact–dimension data models to support production workflow analytics and supply chain reporting.

EcoScan – Web-Based Waste Classification System Using Convolutional Neural Networks (CNN)
Data Science / Data Analyst **Sep 2025 – Des 2025**

- Developed a CNN-based image classification model to identify waste categories with high accuracy.
- Collected, cleaned, and augmented image datasets to improve model generalization and robustness.
- Integrated the trained model into a web application for real-time waste classification.

ORGANISATION EXPERIENCE

Himpunan Mahasiswa Teknik Informatika (HIMIT PENS)
Research and Technology Department Staff **Oct 2024 - Present**

- Coordinated research and learning initiatives to strengthen HIMIT members' technical and academic skills.

Committee of the Mini Class X E-Code, HIMIT PENS 2025
Public Relations Division Committee **Feb 2025 - Mar 2025**

- Organized and facilitated the Mini Class program to enhance PENS students' technical and collaborative skills in UI/UX, frontend–backend development, and presentation readiness.

LKMM Pra-TD 2025 Committee
Liaison Officer Division Committee **Apr 2025**

- Served as a liaison representing the Association of Information Technology and Applied Data Science Students at EEPIS.

Himitopia X Himit Sehat Committee
Liaison Officer Division Committee **Apr 2025 - May 2025**

- Managed public communication and external relations to support the Himitopia X Himit Sehat program.

CERTIFICATION EXPERIENCE

- Grafana | Certificate of Completion (2024)
- Tableau | Certificate of Completion (2024)
- Microsoft Power BI | Certificate of Completion (2024)

HARDSKILLS

Programming & Data Analytics
Python, R, SQL, Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch

Machine Learning & AI
ML Models, Deep Learning (CNN, RNN), Time Series, Model Deployment (CI/CD), Model Evaluation

Data Engineering
ETL (Pentaho), PostgreSQL, MySQL, Data Modeling, API Development, Docker, Git

BI & Visualization
Tableau, Power BI, Grafana

SOFTSKILL

Critical Thinking, Communication Skills, Attention to Detail, Adaptability, Time Management, Continuous Learning, Research Skill, Decision Making

LAMPIRAN PEMOHON 1
TRANSKRIP NILAI



Cetak Transkrip

TRANSKRIP SEMENTARA

Departemen (Department) : Teknik Informatika dan Komputer (Informatics and Computer Engineering)
Program Studi (Study Program) : Program Studi Sarjana Terapan Sains Data Terapan (Bachelor Program of Applied Engineering)
Tanggal Masuk (Entrance Date) : 11-04-2023
Gelar (Degree) : Sarjana Terapan Sains Data Terapan (S.Tr.S.D.T.) (Bachelor of Applied Data Science)

Nama (Name) : Daniar Priambodo
NRP (ID Number) : 3323600009
Tempat, Tanggal Lahir (Place, Date of Birth) : Kediri, 02 Juni 2003
Indeks Prestasi (GPA) : 3.5
Semester Sekarang (Current Semester) : 5

No	Kode (Code)	Mata Kuliah	Subjects	SKS (Credits)	Jam (Hours)	Nilai (Grade)
1	SD041101	Aljabar Linier	Linear Algebra	2	2	AB
2	SD041102	Statistika Dasar	Basic Statistics	2	2	BC
3	SD041103	Praktikum Statistika Dasar	Practice Basic Statistics	1	2	AB
4	SD041104	Pemrograman 1	Programming 1	2	2	A
5	SD041105	Praktikum Pemrograman 1	Practice Programming 1	2	4	A
6	SD041106	Logika dan Algoritma	Logic and Algorithms	2	2	B
7	SD041107	Basis Data	Database	2	2	B
8	SD041108	Praktikum Basis Data	Practice on Database	1	2	AB
9	WI040001	Matematika 1	Mathematics 1	2	2	B
10	WN040001	Agama	Religion	2	2	A
11	KM000001	Kreatifitas Mahasiswa 1	Student Creativity 1	1	3	C
12	SD042101	Pemodelan Statistik Terapan	Applied Statistical Modeling	2	2	AB
13	SD042102	Praktikum Pemodelan Statistik Terapan	Applied Statistical Modeling Practicum	1	2	A
14	SD042103	Pemrograman 2	Programming 2	2	2	AB
15	SD042104	Praktikum Pemrograman 2	Practice on Programming 2	1	2	A
16	SD042105	Kecerdasan Buatan	Artificial Intelligence	2	2	AB
17	SD042106	Praktikum Kecerdasan Buatan	Practice on Artificial Intelligence	1	2	A
18	SD042107	Pemrosesan Data	Data Processing	2	2	AB
19	SD042108	Praktikum Pemrosesan Data	Practice on Data Processing	1	2	AB
20	SD042109	Manajemen Data	Data Management	2	2	A
21	SD042110	Praktikum Manajemen Data	Practice on Data Management	1	2	AB
22	WI040002	Matematika 2	Mathematics 2	2	2	AB
23	WN040003	Kewarganegaraan	Citizenships	2	2	AB
24	SD043101	Matematika 3	Mathematics 3	2	2	A-
25	SD043102	Analisa Statistika Terapan	Applied Statistical Analysis	2	2	AB
26	SD043103	Praktikum Analisa Statistika Terapan	Practice on Applied Statistical Analysis	1	2	A-
27	SD043104	Pemrograman 3	Programming 3	2	2	B
28	SD043105	Praktikum Pemrograman 3	Practice on Programming 3	1	2	BC
29	SD043106	Mesin Pembelajaran	Machine Learning	2	2	AB
30	SD043107	Praktikum Mesin Pembelajaran	Practice on Machine Learning	1	2	AB
31	SD043108	Text Mining	Text Mining	2	2	A-
32	SD043109	Praktikum Text Mining	Practice on Text Mining	1	2	B+
33	SD043110	Eksplorasi dan Visualisasi Data	Data Exploration and Visualization	2	2	A-
34	SD043111	Praktikum Eksplorasi dan Visualisasi Data	Practice on Data Exploration and Visualization	1	2	A-
35	WN04003	Pancasila	Pancasila	2	2	A
36	SD040405	Rekomendasi System	Recommendation System	2	2	A-
37	SD040410	Analisa Bisnis Sains Data	Data Science Business Analysis	2	2	B+
38	SD044101	Data Mining	Data Mining	2	2	A
39	SD044102	Praktikum Data Mining	Practice on Data Mining	1	2	AB
40	SD044103	Teknologi Web Service	Web Technologies	2	2	A-
41	SD044104	Praktikum Teknologi Web Service	Web Technologies Practicum	1	2	AB
42	SD044105	Machine Learning Ops	Machine Learning Operations (MLOps)	2	2	B
43	SD044106	Praktikum Machine Learning Ops	Machine Learning Operations (MLOps) Practicum	1	2	A-
44	SD044107	Infrastruktur dan Manajemen Big Data	Big Data Infrastructure & Management	2	2	A-
45	SD044108	Praktikum Infrastruktur dan Manajemen Big Data	Big Data Infrastructure & Management Practicum	1	2	C
46	SD044109	Data Warehouse	Data Warehouse	2	2	A-
47	SD044110	Praktikum Data Warehouse	Data Warehouse Practicum	1	2	A
48	KP000001	Manajemen Proyek	Project Management	2	2	A-
49	SD045102	Praktikum Neuro Computing	Neuro Computing Practicum	1	2	A
50	SD045104	Workshop Analisis Sosial Media	Social Media Analysis Workshop	2	3	B+
51	SD045107	Technopreneur Sains Data	Data Science Technopreneurship	2	2	A

Catatan Nilai (Grade Explanation)

A	Istimewa (Excellent)	4	BC	Cukup Baik (Sufficient)	2.5
A-		3.75	C	Cukup (Fair)	2
AB	Sangat Baik (Very Good)	3.5	D	Kurang (Poor)	1
B+		3.25	E	Sangat Kurang (Very Poor)	0
B	Baik (Good)	3	Jumlah SKS Total Credits		57 sks

Dosen Wali (Academic Advisor)
Dr. Tita Karlita, S.Kom., M.Kom

Surabaya, 10 Desember 2025
Ketua Program Studi (Head of Program)

Iqbal Uzzin Nadhori, S.Kom., M.T
NIP. 197405052003121002

LAMPIRAN PERMOHONAN 2
CURRICULUM VITAE



ZULFAHMI NAFIIS SUFFI

Address: Jln.Nakula RT 04 / RW 04,Kel. Pucangsewu,Kec. Pacitan,Kab. Pacitan, Jawa Timur
Phone: 0819-3730-6348
Email: zulfahmi9523@gmail.com
Linkedin: linkedin.com/in/zulfahmi-nafiis-suffi

SUMMARY

An undergraduate student in the Applied Data Science D4 program at the Electronic Engineering Polytechnic Institute of Surabaya (EEPIS), with a strong interest in data analysis and data scientist. I have practical experience in handling data and am passionate about turning insights into impactful solutions. I'm a quick learner, adaptable to new environments, and motivated to contribute in collaborative environments.

EDUCATION

Electronic Engineering Polytechnic Institute of Surabaya (EEPIS) **Aug 2023 - Present**
Applied Data Science GPA 3.52/4.00

Senior High School at SMA Negeri 1 Pacitan **2021 - 2023**
Mathematics and Science

PROJECT EXPERIENCE

Langit.In — Climate Forecasting System for East Java
Data Scientist/ Data Analyst **Feb 2025 - Jun 2025**
- Developed climate forecasting models using time-series analysis and machine learning techniques.
- Processed and cleaned multi-year climate datasets to generate accurate forecasting inputs.
- Evaluated model performance using RMSE, MAE, and cross-validation to ensure prediction accuracy.

ERP Manufacturing Data Mart — Production Module Integration
Data Engineer/Data Analyst **May 2025 - Jun 2025**
- Developed a manufacturing data mart integrating the ERP production module using ETL processes in Pentaho.
- Designed and implemented star schema architecture with fact and dimension tables and automated data extraction, transformation, and loading from MySQL ERP databases to support analytics.
- The results were visualized using Power BI to create an interactive dashboard.

Recommendation System — Disease Diagnosis Based On Content-Based filtering Of Early Sytomps
Data Scientist **Apr 2025 - Mar 2025**
- Built a content-based filtering recommendation system to predict possible diseases based on users' early symptoms with high accuracy.
- Implemented a matching algorithm that compares user-input symptoms with symptom similarity scores in the dataset to generate disease recommendations.
- Evaluated the system by testing multiple symptom inputs, producing accurate and responsive disease predictions.

STARK — Analytical Dashboard for Office Supply Management at PENS
Data Analyst **Aug 2025 - Nov 2025**
- Developed an analytical dashboard for managing office supply (ATK) at Politeknik Elektronika Negeri Surabaya (PENS) using machine learning.
- Built predictive models to analyze usage trends and provide data driven procurement recommendations.
- Designed an interactive dashboard to visualize insights for efficient budget planning and decisionmaking.

Public Sentiment Evaluation on Youtube Comments of Purbaya Effect
Data Scientist **Sept 2025 - Oct 2025**
- Built Sentiment analysis on Youtube Comment to Analyzing Pro and Contra of Purbaya effect with NLP
- Designing an interactive dashboard using React.js with sufficient features to analyze the type of emotion in each comment.

EcoScan — Web-Based Waste Classification System Using Convolutional Neural Networks (CNN)
Data Scientist / Data Analyst **Oct 2025 - Nov 2025**
- Developed a CNN-based image classification model to identify waste categories with high accuracy.
- Collected, cleaned, and augmented image datasets to improve model generalization and robustness.
- Integrated the trained model into a web application for real-time waste classification.

ORGANIZATION EXPERIENCE

Minat Bakat Expo PENS (MBEX PENS) 2024 Committee — **Aug 2024**

- Event Division Committee
- Material Presenter of Unit Kegiatan Mahasiswa (UKM)

PENSCOMP (PENS Competition) 2024 Committee — **Oct 2024**

- Forum Supervisor of Supporters Forum
- Match Official
- Health and Safety Division

Association of Information Technology and Applied Data Science Students — **Jan - Apr 2024**

- Campaign Rules and Concept Designer of Election of Leader from Association of Information Technology and Applied Data Science Students

LKMM TD PENS 2025 Committee — **Apr 2025**

- Representatives committee from the Association of Information Technology and Applied Data Science Students at EEPIS
- Committee of Health and Safety Division

Lembaga Minat Bakat (LMB) PENS

Executive Board Department — **Jan 2025 - Nov 2025**

- President of Lembaga Minat Bakat PENS 2025

Student Resource Development Departement — **Feb 2024 - Nov 2024**

- Chairman of Monitoring, Evaluation, and Controlling Agenda 2024
- Junior Staff of Student Resource Development Departement 2024

CERTIFICATION EXPERIENCE

- Learning the Basics of AI Workshop | Dicoding Indonesia (2025)
- Grafana | Certificate of Completion (2024)
- Tableau | Certificate of Completion (2024)
- Microsoft Power BI | Certificate of Completion (2024)

Programming & Data Analytics
Python, R, SQL, Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch

Machine Learning & AI
ML Models, Deep Learning (CNN, RNN), Time Series, Model Deployment (CI/CD), Model Evaluation

Data Engineering
ETL (Pentaho), PostgreSQL, MySQL, Data Modeling, API Development, Docker, Git

BI & Visualization
Tableau, Power BI, Grafana

SOFT SKILL

- Leadership
- Communication Skills
- Critical Thinking
- Adaptability & Continuous Learning
- Presentation Skills
- Research Skill
- Attention to Detail

LAMPIRAN PEMOHON 2
TRANSKRIP NILAI

TRANSKRIP SEMENTARA

Departemen (*Department*) : Teknik Informatika dan Komputer (*Informatics and Computer Engineering*)
Program Studi (*Study Program*) : Program Studi Sarjana Terapan Sains Data Terapan (*Bachelor Program of Applied Engineering*)
Tanggal Masuk (*Entrance Date*) : 04-07-2023
Gelar (*Degree*) : Sarjana Terapan Sains Data Terapan (S.Tr.S.D.T.) (*Bachelor of Applied Data Science*)

Nama (*Name*) : Zulfahmi Nafils Suffi
NRP (*ID Number*) : 3323600019
Tempat, Tanggal Lahir (*Place, Date of Birth*) : Pacitan, 30 Mei 2005
Indeks Prestasi (*GPA*) : 3.5
Semester Sekarang (*Current Semester*) : 5

No	Kode (*Code*)	Mata Kuliah	Subjects	SKS (*Credits*)	Jam (*Hours*)	Nilai (*Grade*)
1	SD041101	Aljabar Linier	*Linear Algebra*	2	2	AB
2	SD041102	Statistika Dasar	*Basic Statistics*	2	2	AB
3	SD041103	Praktikum Statistika Dasar	*Practice Basic Statistics*	1	2	AB
4	SD041104	Pemrograman 1	*Programming 1*	2	2	A
5	SD041105	Praktikum Pemrograman 1	*Practice Programming 1*	2	4	AB
6	SD041106	Logika dan Algoritma	*Logic and Algorithms*	2	2	AB
7	SD041107	Basis Data	*Database*	2	2	A
8	SD041108	Praktikum Basis Data	*Practice on Database*	1	2	AB
9	WI040001	Matematika 1	*Mathematics 1*	2	2	C
10	WN040001	Agama	*Religion*	2	2	A
11	SD042101	Pemodelan Statistik Terapan	*Applied Statistical Modeling*	2	2	B
12	SD042102	Praktikum Pemodelan Statistik Terapan	*Applied Statistical Modeling Practicum*	1	2	A
13	SD042103	Pemrograman 2	*Programming 2*	2	2	A
14	SD042104	Praktikum Pemrograman 2	*Practice on Programming 2*	1	2	AB
15	SD042105	Kecerdasan Buatan	*Artificial Intelligence*	2	2	AB
16	SD042106	Praktikum Kecerdasan Buatan	*Practice on Artificial Intelligence*	1	2	A
17	SD042107	Pemrosesan Data	*Data Processing*	2	2	AB
18	SD042108	Praktikum Pemrosesan Data	*Practice on Data Processing*	1	2	AB
19	SD042109	Manajemen Data	*Data Management*	2	2	A
20	SD042110	Praktikum Manajemen Data	*Practice on Data Management*	1	2	B
21	WI040002	Matematika 2	*Mathematics 2*	2	2	A
22	WN040003	Kewarganegaraan	*Citizenships*	2	2	A
23	SD043101	Matematika 3	*Mathematics 3*	2	2	C
24	SD043102	Analisa Statistika Terapan	*Applied Statistical Analysis*	2	2	AB
25	SD043103	Praktikum Analisa Statistika Terapan	*Practice on Applied Statistical Analysis*	1	2	A-
26	SD043104	Pemrograman 3	*Programming 3*	2	2	BC
27	SD043105	Praktikum Pemrograman 3	*Practice on Programming 3*	1	2	BC
28	SD043106	Mesin Pembelajaran	*Machine Learning*	2	2	AB
29	SD043107	Praktikum Mesin Pembelajaran	*Practice on Machine Learning*	1	2	A-
30	SD043108	Text Mining	*Text Mining*	2	2	A-
31	SD043109	Praktikum Text Mining	*Practice on Text Mining*	1	2	AB
32	SD043110	Eksplorasi dan Visualisasi Data	*Data Exploration and Visualization*	2	2	A-
33	SD043111	Praktikum Eksplorasi dan Visualisasi Data	*Practice on Data Exploration and Visualization*	1	2	A-
34	WN04003	Pancasila	*Pancasila*	2	2	A
35	SD040405	Rekomendasi System	*Recommendation System*	2	2	A
36	SD040410	Analisa Bisnis Sains Data	*Data Science Business Analysis*	2	2	B+
37	SD044101	Data Mining	*Data Mining*	2	2	A
38	SD044102	Praktikum Data Mining	*Practice on Data Mining*	1	2	A-
39	SD044103	Teknologi Web Service	*Web Technologies*	2	2	B+
40	SD044104	Praktikum Teknologi Web Service	*Web Technologies Practicum*	1	2	AB
41	SD044105	Machine Learning Ops	*Machine Learning Operations (MLOps)*	2	2	B+
42	SD044106	Praktikum Machine Learning Ops	*Machine Learning Operations (MLOps) Practicum*	1	2	A-
43	SD044107	Infrastruktur dan Manajemen Big Data	*Big Data Infrastructure & Management*	2	2	B+
44	SD044109	Data Warehouse	*Data Warehouse*	2	2	A-
45	SD044110	Praktikum Data Warehouse	*Data Warehouse Practicum*	1	2	A
46	KP000001	Manajemen Proyek	*Project Management*	2	2	A
47	KP000005	English for developing vocabulary	*English for developing vocabulary*	2	2	AB
48	SD045102	Praktikum Neuro Computing	*Neuro Computing Practicum*	1	2	AB
49	SD045104	Workshop Analisis Sosial Media	*Social Media Analysis Workshop*	2	3	AB
50	SD045107	Technopreneur Sains Data	*Data Science Technopreneurship*	2	2	A

Catatan Nilai (*Grade Explanation*)

A	Istimewa (*Excellent*)	4	BC	Cukup Baik (*Sufficient*)	2.5
A-		3.75	C	Cukup (*Fair*)	2
AB		3.5	D	Kurang (*Poor*)	1
B+	Sangat Baik (*Very Good*)	3.25	E	Sangat Kurang (*Very Poor*)	0
B	Baik (*Good*)	3			

Jumlah SKS Total Credits : 56 sks

Dosen Wali (*Academic Advisor*)

Dr. Tita Karlita, S.Kom., M.Kom

Surabaya, 10 Desember 2025
Ketua Program Studi (*Head of Program*)

Isbat Uzzin Nadhori, S. Kom., M.T
NIP. 197405052003121002

LAMPIRAN PEMOHON 3
CURRICULUM VITAE



Zacky Ainul Mustofa

Address : DSN Perning 007/001 Perning, Jatikalen, Nganjuk
Email : zackymustafa7@gmail.com
Linkedin: https://www.linkedin.com/in/zacky-ainul-mustofa/
Phone : +62 877 6104 7433

Profile

Active 6th Semester students of the Applied Data Science Study Program, Surabaya State Electronics Polytechnic who have an interest and desire to develop skills in the fields of *Data Analyst, Data Engineering and Data Scientist*.

Education

SMA Progresif Bumi Sholawat Sidoarjo **2020 - 2023**
Mathematics And Science

Politeknik Elektronika Negeri Surabaya **2023 - Now**
Applied Data Science GPA 3.3/4.00

Project

Langit.In — Climate Forecasting System for East Java **Feb 2025 - Jun 2025**
Data Science / Data Analyst

- Developed climate forecasting models using time-series analysis and machine learning techniques.
- Processed and cleaned multi-year climate datasets to generate accurate forecasting inputs.
- Evaluated model performance using RMSE, MAE, and cross-validation to ensure prediction accuracy.

ERP Manufacturing Data Mart — Production Module Integration **May 2025 - Jun 2025**
Data Engineer

- Developed and managed ETL pipelines using Pentaho Data Integration (PDI) to integrate production module data into the Supply Chain Data Mart.
- Performed data extraction from PostgreSQL and MySQL ERP sources, followed by transformation and loading into centralized analytical tables.
- Designed optimized fact–dimension data models to support production workflow analytics and supply chain reporting.

STARK — Analytical Dashboard for Office Supply Management at PENS **Aug 2025 - Nov 2025**
Data Analyst

- Developed an analytical dashboard for managing office supply (ATK) at Politeknik Elektronika Negeri Surabaya (PENS) using machine learning.
- Built predictive models to analyze usage trends and provide data driven procurement recommendations.
- Designed an interactive dashboard to visualize insights for efficient budget planning and decisionmaking.

Public Sentiment Evaluation on Youtube of Purbaya Effect **Sept 2025 - Oct 2025**
Data Scientist

- Built Sentiment analysis on Youtube Comment to Analyzing Pro and Contra of Purbaya effect with NLP
- Designing an interactive dashboard using React.js with sufficient features to analyze the type of emotion in each comment.

EcoScan — App for Classification Trash **Oct 2025 - Nov 2025**
Data Scientist / Data Analyst

- Developed a CNN-based image classification model to identify waste categories with high accuracy.
- Collected, cleaned, and augmented image datasets to improve model generalization and robustness.
- Integrated the trained model into a web application for real-time waste classification.

Organizational and Committee Experience

LKMM Pra-TD PENS **2023**
Participants
- Participated in the Pre-TD Student Management Skills Training activity

LKMM TD PENS **2024**
Participants
- ·Participated in the TD Student Management Skills Training activity

Himpunan Mahasiswa TeknikInformatika **2024 - 2025**
Junior Staff of the Department of Internal Affairs
- Implements the work program for internal family activities of Informatics Engineering students
- Participates in activities organized by several other departments

OPP Dynamic 2024 HIMIT PENS **2025**
Instructor Committee (IC)
- ·Becoming a senior group mentor during the study program introduction period

OB Pendewasaan Amanah 2024 **2025**
Steering Committee (SC)
- Becoming the conceptor of the 2024 Amanah Maturation OB Activity

HIMITOPIA 2025 **2025**
Person in Charge
- Person in charge of HIMITOPIA event

SAMWIS HIMIT 2025 **2025**
Person in Charge
- Person in charge of the 2025 HIMIT Graduation Welcome Event

Portfolio

- https://zackymustafa.vercel.app
- https://github.com/Zacky-zem

Certification Experience

- Grafana | Certificate of Completion (2024)
- Tableau | Certificate of Completion (2024)
- Microsoft Power BI | Certificate of Completion (2024)

Skills

Soft Skills: Communication, Teamwork, Adaptabilit

Microsoft Office: Ms. Word, Ms. Excel, Ms. Powerpoint

Software Tools: R Studio, VS Code, Power BI, Tableau, Grafana, Pentaho

Design and Editing: Canva, Capcut

Programming & Data Analytics: Python, R, SQL, Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch

ML & AI: ML Models, Deep Learning (CNN, RNN), Time Series, Model Deployment (CI/CD), Model Evaluation

Data Engineering: ETL (Pentaho), PostgreSQL, MySQL, Data Modeling, API Development, Docker, Git

LAMPIRAN PEMOHON 3
TRANSKRIP NILAI

Cetak Transkrip

TRANSKRIP SEMENTARA

Departemen (*Department*) : Teknik Informatika dan Komputer (*Informatics and Computer Engineering*)
Program Studi (*Study Program*) : Program Studi Sarjana Terapan Sains Data Terapan (*Bachelor Program of Applied Engineering*)
Tanggal Masuk (*Entrance Date*) : 10-04-2023
Gelar (*Degree*) : Sarjana Terapan Sains Data Terapan (S.Tr.S.D.T.) (*Bachelor of Applied Data Science*)

Nama (*Name*) : Zacky Ainul Mustofa
NRP (*ID Number*) : 3323600007
Tempat, Tanggal Lahir (*Place, Date of Birth*) : NGANJUK, 06 Desember 2004
Indeks Prestasi (*GPA*) : 3.3
Semester Sekarang (*Current Semester*) : 5

No	Kode (*Code*)	Mata Kuliah	Subjects	SKS (*Credits*)	Jam (*Hours*)	Nilai (*Grade*)
1	SD041101	Aljabar Linier	*Linear Algebra*	2	2	AB
2	SD041102	Statistika Dasar	*Basic Statistics*	2	2	B
3	SD041103	Praktikum Statistika Dasar	*Practice Basic Statistics*	1	2	AB
4	SD041104	Pemrograman 1	*Programming 1*	2	2	BC
5	SD041105	Praktikum Pemrograman 1	*Practice Programming 1*	2	4	B
6	SD041106	Logika dan Algoritma	*Logic and Algorithms*	2	2	BC
7	SD041107	Basis Data	*Database*	2	2	B
8	SD041108	Praktikum Basis Data	*Practice on Database*	1	2	AB
9	WI040001	Matematika 1	*Mathematics 1*	2	2	C
10	WN040001	Agama	*Religion*	2	2	A
11	SD042101	Pemodelan Statistik Terapan	*Applied Statistical Modeling*	2	2	AB
12	SD042102	Praktikum Pemodelan Statistik Terapan	*Applied Statistical Modeling Practicum*	1	2	AB
13	SD042104	Praktikum Pemrograman 2	*Practice on Programming 2*	1	2	AB
14	SD042105	Kecerdasan Buatan	*Artificial Intelligence*	2	2	AB
15	SD042106	Praktikum Kecerdasan Buatan	*Practice on Artificial Intelligence*	1	2	AB
16	SD042107	Pemrosesan Data	*Data Processing*	2	2	AB
17	SD042108	Praktikum Pemrosesan Data	*Practice on Data Processing*	1	2	AB
18	SD042109	Manajemen Data	*Data Management*	2	2	A
19	SD042110	Praktikum Manajemen Data	*Practice on Data Management*	1	2	B
20	WI040002	Matematika 2	*Mathematics 2*	2	2	AB
21	WN040003	Kewarganegaraan	*Citizenships*	2	2	AB
22	SD042103	Pemrograman 2	*Programming 2*	2	2	BC
23	SD043101	Matematika 3	*Mathematics 3*	2	2	AB
24	SD043102	Analisa Statistika Terapan	*Applied Statistical Analysis*	2	2	B+
25	SD043103	Praktikum Analisa Statistika Terapan	*Practice on Applied Statistical Analysis*	1	2	AB
26	SD043104	Pemrograman 3	*Programming 3*	2	2	B
27	SD043105	Praktikum Pemrograman 3	*Practice on Programming 3*	1	2	BC
28	SD043106	Mesin Pembelajaran	*Machine Learning*	2	2	AB
29	SD043107	Praktikum Mesin Pembelajaran	*Practice on Machine Learning*	1	2	AB
30	SD043108	Text Mining	*Text Mining*	2	2	A-
31	SD043109	Praktikum Text Mining	*Practice on Text Mining*	1	2	B+
32	SD043110	Eksplorasi dan Visualisasi Data	*Data Exploration and Visualization*	2	2	A-
33	SD043111	Praktikum Eksplorasi dan Visualisasi Data	*Practice on Data Exploration and Visualization*	1	2	A-
34	WN04003	Pancasila	*Pancasila*	2	2	A
35	SD040405	Rekomendasi System	*Recommendation System*	2	2	AB
36	SD040410	Analisa Bisnis Sains Data	*Data Science Business Analysis*	2	2	B+
37	SD044101	Data Mining	*Data Mining*	2	2	A-
38	SD044102	Praktikum Data Mining	*Practice on Data Mining*	1	2	AB
39	SD044103	Teknologi Web Service	*Web Technologies*	2	2	AB
40	SD044104	Praktikum Teknologi Web Service	*Web Technologies Practicum*	1	2	AB
41	SD044105	Machine Learning Ops	*Machine Learning Operations (MLOps)*	2	2	B+
42	SD044106	Praktikum Machine Learning Ops	*Machine Learning Operations (MLOps) Practicum*	1	2	A-
43	SD044107	Infrastruktur dan Manajemen Big Data	*Big Data Infrastructure & Management*	2	2	AB
44	SD044108	Praktikum Infrastruktur dan Manajemen Big Data	*Big Data Infrastructure & Management Practicum*	1	2	C
45	SD044109	Data Warehouse	*Data Warehouse*	2	2	AB
46	SD044110	Praktikum Data Warehouse	*Data Warehouse Practicum*	1	2	AB
47	KP000001	Manajemen Proyek	*Project Management*	2	2	A-
48	SD045102	Praktikum Neuro Computing	*Neuro Computing Practicum*	1	2	AB
49	SD045104	Workshop Analisis Sosial Media	*Social Media Analysis Workshop*	2	3	B+
50	SD045107	Technopreneur Sains Data	*Data Science Technopreneurship*	2	2	A

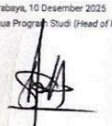
Catatan Nilai (*Grade Explanation*)

A	Istimewa (*Excellent*)	4	BC	Cukup Baik (*Sufficient*)	2.5
A-		3.75	C	Cukup (*Fair*)	2
AB	Sangat Baik (*Very Good*)	3.5	D	Kurang (*Poor*)	1
B+		3.25	E	Sangat Kurang (*Very Poor*)	0
B	Baik (*Good*)	3	Jumlah SKS Total Credits		56 sks

Dosen Wali (*Academic Advisor*)

Dr. Tita Karlita, S.Kom., M.Kom

Surabaya, 10 Desember 2025
Ketua Program Studi (*Head of Program*)



Isbat Uzzin Nadhori, S.Kom., M.T
NIP. 197405052003121002

LAMPIRAN PEMOHON 4
CURRICULUM VITAE



MUHAMMAD ZAKI ZAIN FANURUDDIN PUTRA

Address: Jl. Ngagel Dadi 1G No.12
Phone: 0812-1756-817
Email: zakizainn18@gmail.com
Linkedin: linkedin.com/in/muhammadzakizainfp/

SUMMARY

Undergraduate student in Applied Data Science (5th semester) at Electronic Engineering Polytechnic Institute of Surabaya (EEPIS), with a keen interest in Data Analysis and Data Engineering. Experienced in applying data science techniques, data processing, and analytical tools. A fast learner with strong adaptability and proven ability to collaborate effectively in diverse team environments.

EDUCATION

Bachelor of Applied Data Science (Undergraduate) **2023 - Present**
Electronic Engineering Polytechnic Institute of Surabaya
- Currently in 5th semester.
- Relevant coursework: Data Analysis, Data Engineering, Machine Learning, Database Systems.

Senior High School (Mathematics and Science Major) **2020 - 2023**
State Senior High School 1 Surabaya (SMA Negeri 1 Surabaya)
- Majoring in Mathematics and Science (MIPA).
- Solid background in mathematics, physics.

PROJECT

FAZtream **Feb 2025 - June 2025**
- Built the project's foundational movie dataset by scraping multiple web sources and enriching missing information through the TMDB API, ensuring complete and reliable data for model training and user consumption.
- Processed, cleaned, and structured large volumes of movie metadata, transforming unorganized raw data into an analysis-ready dataset that strengthened the accuracy and relevance of the recommendation system.
- Designed the ERD and deployed the database through FastAPI, enabling efficient data delivery, seamless backend integration, and a scalable infrastructure for future feature expansion.

Pantau.In **Aug 2025 - Nov 2025**
- Conducted data cleaning, validation, and labeling to correct inconsistencies and ensure a reliable foundation for further analysis, significantly improving dataset quality and model readiness.
- Performed clustering and comparative data analysis to uncover behavioral patterns and meaningful insights, supporting more informed decision-making within the project.
- Led data migration, transformation, visualization, and reporting activities—delivering clear, insight-driven dashboards while maintaining strong data quality assurance throughout the workflow.

ORGANIZATION EXPERIENCE

LKMM TD HIMIT PENS **2025**
Instructor Committee (IC)
- I assisted a student group during soft-skills and leadership training sessions.

Lembaga Minat Bakat PENS **2024 - Present**
- Head of Student Resource Development Departement
- Chairman of Suksesi LMB PENS 2024
- Junior Staff of Student Resource Development Departement 2024

PORSENI POLITEKNIK XIV 2024 **July 2024**
Team Official
- Supporting athletes throughout both the pre-event and competition phases.

Technogear PENS 2024 **August 2024**
Liaison Officer (LO)
- I assisted and guided new students throughout the New Student Orientation program.

PENSCOMP 2024 **September 2024**
Member of Event Division
- I was responsible for organizing a futsal competition and coordinating HIMA events, handling event preparation, participant management, and ensuring smooth execution.

OPP Dynamic 2024 HIMIT PENS **2025**
Instructor Committee (IC)
- I guided a group of new students during the study program introduction.

OB Dynamic 2024 HIMIT PENS **2025**
Instructor Committee (IC)
- I supported and guided a group of new students during the department's orientation.

WORK EXPERIENCE

Freelance - Pentamoo **2024**
Event Crew

PORTFOLIO

mzakizainfp.vercel.app

https://github.com/zakizainn/

https://pantauinsdgswates.vercel.app/

SKILLS

- **Programming Language:** Python, SQL, R, Javascript
- **Languages:** Indonesia, English.
- **Certifications:** Advanced Data Visualization with Tableau, Microsoft Power BI, and Grafana.
- **Soft Skills:** Teamwork, Communication, Adaptability

LAMPIRAN PEMOHON 4
TRANSKRIP NILAI

Cetak Transkrip

TRANSKRIP SEMENTARA

Departemen (*Department*) : Teknik Informatika dan Komputer (*Informatics and Computer Engineering*)
Program Studi (*Study Program*) : Program Studi Sarjana Terapan Sains Data Terapan (*Bachelor Program of Applied Engineering*)
Tanggal Masuk (*Entrance Date*) : 10-04-2023
Gelar (*Degree*) : Sarjana Terapan Sains Data Terapan (S.Tr.S.D.T.) (*Bachelor of Applied Data Science*)

Nama (*Name*) : Muhammad Zaki Zain Fanuruddin Putra
NRP (*ID Number*) : 3323600006
Tempat, Tanggal Lahir (*Place, Date of Birth*) : SURABAYA, 01 Agustus 2005
Indeks Prestasi (*GPA*) : 3.34
Semester Sekarang (*Current Semester*) : 5

No	Kode (*Code*)	Mata Kuliah	Subjects	SKS (*Credits*)	Jam (*Hours*)	Nilai (*Grade*)
1	SD041101	Aljabar Linier	*Linear Algebra*	2	2	AB
2	SD041102	Statistika Dasar	*Basic Statistics*	2	2	AB
3	SD041103	Praktikum Statistika Dasar	*Practice Basic Statistics*	1	2	AB
4	SD041104	Pemrograman 1	*Programming 1*	2	2	BC
5	SD041105	Praktikum Pemrograman 1	*Practice Programming 1*	2	4	AB
6	SD041106	Logika dan Algoritma	*Logic and Algorithms*	2	2	B
7	SD041107	Basis Data	*Database*	2	2	AB
8	SD041108	Praktikum Basis Data	*Practice on Database*	1	2	A
9	WI040001	Matematika 1	*Mathematics 1*	2	2	C
10	WN040001	Agama	*Religion*	2	2	A
11	SD042101	Pemodelan Statistik Terapan	*Applied Statistical Modeling*	2	2	B
12	SD042102	Praktikum Pemodelan Statistik Terapan	*Applied Statistical Modeling Practicum*	1	2	AB
13	SD042103	Pemrograman 2	*Programming 2*	2	2	C
14	SD042104	Praktikum Pemrograman 2	*Practice on Programming 2*	1	2	AB
15	SD042105	Kecerdasan Buatan	*Artificial intelligence*	2	2	AB
16	SD042106	Praktikum Kecerdasan Buatan	*Practice on Artificial Intelligence*	1	2	A
17	SD042107	Pemrosesan Data	*Data Processing*	2	2	AB
18	SD042108	Praktikum Pemrosesan Data	*Practice on Data Processing*	1	2	AB
19	SD042109	Manajemen Data	*Data Management*	2	2	A
20	SD042110	Praktikum Manajemen Data	*Practice on Data Management*	1	2	B
21	WI040002	Matematika 2	*Mathematics 2*	2	2	A
22	WN040003	Kewarganegaraan	*Citizenships*	2	2	C
23	SD043101	Matematika 3	*Mathematics 3*	2	2	A-
24	SD043102	Analisa Statistika Terapan	*Applied Statistical Analysis*	2	2	B+
25	SD043103	Praktikum Analisa Statistika Terapan	*Practice on Applied Statistical Analysis*	1	2	AB
26	SD043104	Pemrograman 3	*Programming 3*	2	2	B+
27	SD043105	Praktikum Pemrograman 3	*Practice on Programming 3*	1	2	BC
28	SD043106	Mesin Pembelajaran	*Machine Learning*	2	2	AB
29	SD043107	Praktikum Mesin Pembelajaran	*Practice on Machine Learning*	1	2	A-
30	SD043108	Text Mining	*Text Mining*	2	2	A-
31	SD043109	Praktikum Text Mining	*Practice on Text Mining*	1	2	B+
32	SD043110	Eksplorasi dan Visualisasi Data	*Data Exploration and Visualization*	2	2	A-
33	SD043111	Praktikum Eksplorasi dan Visualisasi Data	*Practice on Data Exploration and Visualization*	1	2	A-
34	WN04003	Pancasila	*Pancasila*	2	2	A
35	SD040405	Rekomendasi System	*Recommendation System*	2	2	AB
36	SD040410	Analisa Bisnis Sains Data	*Data Science Business Analysis*	2	2	B+
37	SD044101	Data Mining	*Data Mining*	2	2	A
38	SD044102	Praktikum Data Mining	*Practice on Data Mining*	1	2	AB
39	SD044103	Teknologi Web Service	*Web Technologies*	2	2	B+
40	SD044104	Praktikum Teknologi Web Service	*Web Technologies Practicum*	1	2	AB
41	SD044105	Machine Learning Ops	*Machine Learning Operations (MLOps)*	2	2	AB
42	SD044106	Praktikum Machine Learning Ops	*Machine Learning Operations (MLOps) Practicum*	1	2	B+
43	SD044107	Infrastruktur dan Manajemen Big Data	*Big Data Infrastructure & Management*	2	2	A-
44	SD044108	Praktikum Infrastruktur dan Manajemen Big Data	*Big Data Infrastructure & Management Practicum*	1	2	C
45	SD044109	Data Warehouse	*Data Warehouse*	2	2	A-
46	SD044110	Praktikum Data Warehouse	*Data Warehouse Practicum*	1	2	A
47	KP000001	Manajemen Proyek	*Project Management*	2	2	A-
48	SD045102	Praktikum Neuro Computing	*Neuro Computing Practicum*	1	2	AB
49	SD045104	Workshop Analisis Sosial Media	*Social Media Analysis Workshop*	2	3	B+
50	SD045107	Technopreneur Sains Data	*Data Science Technopreneurship*	2	2	AB

Catatan Nilai (*Grade Explanation*)

A	Istimewa (*Excellent*)	4	BC	Cukup Baik (*Sufficient*)	2.5
A-		3.75	C	Cukup (*Fair*)	2
AB	Sangat Baik (*Very Good*)	3.5	D	Kurang (*Poor*)	1
B+		3.25	E	Sangat Kurang (*Very Poor*)	0
B	Baik (*Good*)	3	Jumlah SKS Total Credits		56 sks

Dosen Wali (*Academic Advisor*)

Dr. Tita Karlita, S.Kom., M.Kom

Surabaya, 10 Desember 2025
Ketua Program Studi (*Head of Program*)

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